

Supplementary Materials of AssembledHOI

Outline

In this work, we present AssembledHOI, a novel framework that achieves efficient and accurate 3D HOI synthesis through structured assembly of Gaussian priors for humans and objects. This supplementary document provides additional experimental details and results to support our approach. The content is organized as follows:

- Sec. **A** includes additional experiments that further validate the effectiveness and superiority of AssembledHOI, including qualitative comparison with baseline methods, quantitative evaluation with other approaches, user study (human perceptual studies) and benchmarks such as geometry sampling time.
- Sec. **B** details the technical implementation of our method, covering prompt formulation, evaluation details, baseline configuration, training details, and the GPT-4V-guided interaction part recognition pipeline, including its robustness fallback mechanism.
- Sec. **C** discusses the current limitations of our approach and outlines directions for future work.

A. Experiments

Comparison with Baseline Methods. We provide additional visual comparisons with baseline methods, as shown in Fig.1. Our approach achieves superior performance compared to the baseline methods in terms of both geometric consistency and texture fidelity. As illustrated in Fig. 2a, more samples generated by our AssembledHOI under diverse prompts demonstrate not only strong alignment with the input prompts, but also high-fidelity details and accurate spatial articulation. Moreover, we have also demonstrated scenarios involving multi-object interaction in Fig. 2b, showcasing the strong compositional generalization ability of our framework.

Comparison with Other Approaches. Recent works such as DreamHOI [16] (**Code released, but texture is limited and training time is over 11 hours**) and InteractAnything [15] (**No code released**) employ similar explicit modeling approaches for zero-shot Human-Object interaction generation. Nevertheless, their methods still differ significantly from our work. DreamHOI [16] fixes the geometry of the object during human pose refinement, leading to misaligned contacts and penetration artifacts due to the lack of object adaptability and geometric interaction priors. On the other hand, InteractAnything [15] leverages LLM-guided semantic feedback and 2D affordance parsing for text-geometry



Fig. 1: Qualitative comparisons between our method and DreamFusion [10], Magic3D [8], TextMesh [12] and InterFusion [2]. Our method is measured on RTX 3090 and uses 17GB GPU memory.

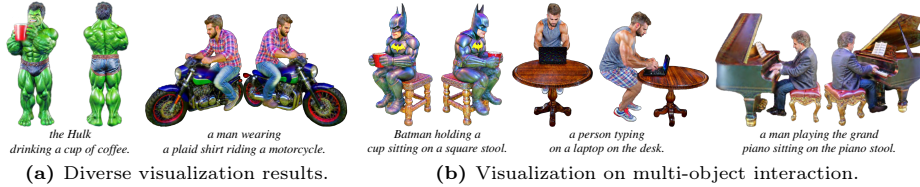


Fig. 2: Visualization results on different text prompts and multi-object interaction.

alignment but lacks of texture fidelity. Our method overcomes both limitations: we separately optimize the 6DoF pose, scale, and shape of both the human and the object, enabling mutual adaptation. Furthermore, we introduce a unified interaction-aware pipeline, comprising Adaptive Anchoring Selection (AAS), Part-Aware Local Refinement (PALR) and Part-Guided Textural Enhancement (PGTE), to explicitly refine contact regions and co-optimize geometry and texture at the part level. Since DreamHOI requires additional clothing texture maps to display different textures, we used the default texture in the main text for visualization comparison to ensure fairness. During experiments, we also employed the same texture description in the prompt. Due to the lack of code and texture, we adopt the CLIP score in their papers [15, 16]. We can notice that the score is obviously lower than our method.

Comparison with Training-based methods. We compared our zero-shot approach with state-of-the-art 3D generation models **trained on large-scale 3D assets**. Taking the representative Hunyuan3D [6] as an example, we conducted experiments on the same test sample set as the baseline methods, as

Table 1: More quantitative comparison with baseline methods.

Method	DreamFusion	Magic3D	TextMesh	InterFusion	DreamHOI	Ours
FID _{CLIP} (%) ↓	68.4	70.0	69.8	63.7	65.2	62.3
R-Precision(%) ↑	68.8	73.8	47.5	83.6	76.4	85.7

**Fig. 3:** Qualitative and quantitative comparisons with Hunyuan3D $v3.1$ [6]. The other examples of ours are presented in Fig.3 and Fig.5 in the main text.

illustrated in Fig. 3. The results indicate that our method consuming a lower computational resource achieves a CLIP score of 0.3835, which is comparable to Hunyuan3D (0.4049), although a certain gap persists in visual quality. We shall endeavor to narrow this gap in future work.

Comparison with AvatarCLIP [4]. It is important to note that AvatarCLIP’s close-up optimization differs significantly from our approach. AvatarCLIP employs NeuS [13] implicit representation for rendering, where the generated 2D images undergo localized detail enhancement for face and back regions guided by fixed prompt prefixes (“the face of”, “the back of”). In contrast, our approach employs explicit representation 3DGS [5] for rendering. The human pose is divided into **12** interaction regions, and through region masks, we can perform localized optimization on any specific area without requiring fixed text prompts. An example of a 2D region projection image is shown in Fig. 5.

Additional quantitative comparisons. We incorporate CLIP R-Precision and FID_{CLIP} into our evaluation metrics, as shown in Tab. 1. The CLIP R-Precision metric, from the text-to-image generation literature, is the retrieval accuracy with which CLIP retrieves the matching caption among rendered images, evaluates the relevance of the retrieved 3D models to the textual queries. FID_{CLIP} assesses the visual fidelity of our generated scenes within the CLIP feature space.

User Study. Although automated metrics such as CLIP score and GPT-4V select provide convenient proxies for evaluating text-to-3D generation, they suffer from inherent limitations. The CLIP score mainly assesses semantic similarity between text and rendered views, yet it does not account for physical plausibility, motion coherence, or fine-grained interaction dynamics. Similarly, while GPT-4V select is capable of evaluating multi-view consistency and structural realism, it remains unreliable in assessing finer aspects such as texture quality, pose naturalness, or the subtle coordination between humans and objects. Crucially, these evaluation metrics fail to capture human perception judgments of “realism” or “naturalness”, which are essential for creating immersive 3D content.

Table 2: User Study Results on SA, PP, IN, TF and OP (10-point Likert Scale). SA denotes Semantic Alignment, PP denotes Physical Plausibility, IN denotes Interaction Naturalness, TF denotes Texture Fidelity and OP denotes Overall Preference.

Method	DreamFusion	Magic3D	TextMesh	InterFusion	DreamHOI	Ours
SA $\uparrow$	6.9 $\pm$ 0.5	6.7 $\pm$ 0.8	6.4 $\pm$ 0.9	7.0 $\pm$ 1.0	7.2 $\pm$ 0.7	8.9$\pm$0.5*
PP $\uparrow$	6.7 $\pm$ 0.9	6.5 $\pm$ 1.0	6.2 $\pm$ 1.2	7.0 $\pm$ 1.3	6.8 $\pm$ 1.4	8.6$\pm$0.6*
IN $\uparrow$	6.6 $\pm$ 0.6	6.3 $\pm$ 0.8	6.1 $\pm$ 1.1	6.5 $\pm$ 1.5	6.8 $\pm$ 1.2	8.5$\pm$0.8*
TF $\uparrow$	6.4 $\pm$ 0.4	6.5 $\pm$ 0.6	6.6 $\pm$ 0.5	6.8 $\pm$ 0.4	6.0 $\pm$ 1.3	9.0$\pm$0.3*
OP $\uparrow$	6.8 $\pm$ 0.5	6.6 $\pm$ 0.6	6.4 $\pm$ 0.7	7.1 $\pm$ 0.5	7.0 $\pm$ 0.6	8.9$\pm$0.3*

* denotes statistical significance at $p < 0.01$ (two-tailed paired t-test) compared to all other methods.

Table 3: Quantitative comparison with previous methods.

Method	GaussianDreamer [14]	Ours
Geometry Sampling Time $\downarrow$	10 s	5 s

To address this gap, we conducted a controlled user study with **30** participants who had basic familiarity with 3D visualization. Each participant evaluates **10** distinct human-object interaction scenes, with each scene containing multiple video renderings (MP4) produced by competing methods. The videos were presented in randomized order to minimize ordering bias, and participants were blinded to the method identities. For each video, participants rated five dimensions on a 10-point scale: *Semantic Alignment* (consistency with the input prompt), *Physical Plausibility* (adherence to real-world physics), *Interaction Naturalness* (fluidity and intuitiveness of human-object contact), *Texture Fidelity* (visual quality), and *Overall Preference*. Mean $\pm$ standard deviation values are reported across participants. Statistical significance was assessed using two-tailed paired t-tests with Bonferroni correction ($\alpha = 0.01$).

The study protocol was reviewed and exempted by our institutional ethics board. Tab. 2 offers a more comprehensive and perception-based evaluation, highlighting substantial discrepancies between automated metrics and human judgments, especially in complex interaction scenarios that require contextual reasoning and embodied intuition.

Geometry Sampling Time Comparison. Geometry Sampling method from GaussianDreamer [14] suffers from prohibitive computational latency due to sequential nearest-neighbor queries and lack of parallelization, limiting scalability in high-resolution 3D initialization. To address this, we implement a batched and parallelized sampling strategy. Specifically, we use a SciPy-based cKDTree [1] to perform nearest-neighbor lookups in batches, which allows multi-threaded execution. As shown in Tab. 3, our approach reduces the sampling time by approx-

Input Prompt Template (JSON)
<pre> "sitting on a sofa":{ "h": "Iron man", "o": "a brown leather sofa", "i": "Iron man sitting on a brown leather sofa, 8k, ultra-detailed" }, "carrying a backpack": { "h": "a man", "o": "a backpack with black straps", "i": "a man in a hoodie, camouflage jacket, joggers and trainers carrying a backpack, 8k, ultra-detailed" }, "jumping a skateboard": { "h": "the spider-man", "o": "a skateboard with wheels", "i": "the spider-man standing on a skateboard, 8k, ultra-detailed" }, /... </pre>

Fig. 4: Input prompt format for text-to-3D human-object interaction generation. Each scene specifies the action, human (h), object (o), and a unified description (i) that combines both entities with high-resolution guidance using a single suffix (“8k, ultra-detailed”).

imately half compared to GaussianDreamer [14] for an entity, demonstrating a practical speedup without compromising initialization quality.

B. Implementation Details

Prompting. Our method takes a structured text prompt as input, which consists of four components: (1) an *action* describing the interaction (*e.g.*, “carrying a backpack”), the human description (h), the object description (o) and a unified prompt (i) that seamlessly combines both entities into a single natural-language sentence enriched with high-fidelity rendering cues (*e.g.*, “8k, ultra-detailed”). This design ensures semantic consistency between the human and the object while providing strong guidance for geometry and texture synthesis. Notably, only the unified prompt i is used during the 3D generation stage; h and o are used primarily for gaussian prior generation in our pipeline. The prompt configuration template is shown in Fig. 4.

Evaluation Metrics Details. For the evaluation sample design, we adopted an InterFusion [2] assessment methodology, selecting 245 (including 61 in InterFusion) unique and diverse prompts to ensure balanced representation across human categories (*e.g.*, soldiers, teachers), clothing styles (*e.g.*, hoodies, jackets), objects (*e.g.*, motorcycles, shields) and interaction modes (*e.g.*, riding, holding). This selection covered 13 distinct interaction types, involving varied contact areas across the full body. For the CLIP score metric, we utilized the official example-based evaluation protocol to assess the consistency between the input

Table 4: Configuration settings comparison with baseline methods.

Method	DreamFusion	Magic3D	TextMesh	InterFusion	DreamHOI	Ours	Ours
Batch Size	8	8	4	1	8	4	4
Resolution	512*512	512*512	512*512	512*512	512*512	1024*1024	1024*1024
Epochs	10000	10000	10000	10000	10000	2400	2400
GPU	vGPU-32GB	vGPU-32GB	vGPU-32GB	vGPU-32GB	vGPU-32GB	vGPU-32GB	RTX 3090
Memory	10GB	15GB	15GB	29GB	31GB	17GB	17GB
Time ↓	1.8 hours	2.0 hours	1.5 hours	4.5 hours	11.5 hours	30 minutes	25 minutes

text prompts and the generated 120 multi-view rendered images, computing the average score accordingly. The CLIP variant used was *clip-ViT-B/16*. For the GPT-4V select metric, we instructed the model to select the index value from all generated results that best aligns with the input text in terms of physical integrity and interactional plausibility. The prompt template used for GPT-4V was: “*My input text prompt is..., please select the index value from the given images that best matches the input text in terms of physical integrity and interactional plausibility.*” It should be noted that contextual examples were not provided as guidance during the evaluation, and the order of the generated results was randomly shuffled. Given the diversity of the samples, repeating the evaluation with multiple random seeds was deemed unnecessary. Subsequently, **User Study** was conducted to manually verify the consistency of GPT-4V select, ensuring a comprehensive and reliable assessment. The evaluation results are summarized in the main text. We recommend that readers use GPT-4V [9] to evaluate the results presented in this work, as it can provide more detailed and insightful feedback. The timing metric is discussed in detail in the **Baseline Configuration** section, where the setup for each method is fully specified. These measurements were conducted personally by the author, ensuring consistency with the reported findings and eliminating potential discrepancies.

Baseline Configuration. To ensure a fair comparison, all baseline methods were evaluated under identical experimental conditions. Since the 24GB VRAM of the RTX 3090 graphics card was insufficient to meet the operational requirements of Interfusion [2] and DreamHOI [16], we uniformly utilized the 32GB vGPU provided by the Autodl platform for model evaluation. Nevertheless, as stated in the main text, our proposed method was also tested on the RTX 3090 to validate its efficiency under more constrained hardware conditions. For each baseline, Tab. 4 summarizes key configuration settings including batch size, rendering resolution, training epochs, GPU model, GPU memory usage and training time. To ensure the reliability of timing measurements, each baseline was evaluated over **10** independent runs with diverse text prompts, and the results were averaged. This approach ensures that observed improvements in speed are attributable to algorithmic advances in our method rather than fluctuations in implementation or hardware resource allocation.

Training Details. We provide additional training specifics that were omitted from the main paper due to space constraints. The full parameter settings are listed below. For the 2D diffusion model, we use *stabilityai/stable-diffusion-2-1-base* [11], with a guidance scale of 100. For optimization stability, we initialized all 6 degrees of freedom (6DOF) for human and object transformations at a learning rate of 0.01. The learning rate follows a schedule that decays to 0.001 after 1200 iterations and remains constant thereafter. To further stabilize geometric optimization, adaptive anchors were employed, fixing viewpoints every 100 iterations up to 1200 iterations. During the initial 1200 iterations, the time steps of adding noise were set as $(0.58 - 0.78)$ to facilitate geometry optimizations. From iteration 1200 to 1700, the time step range was adjusted to $(0.02 - 0.98)$, ensuring rich texture variations. After 1700 iterations, this range was narrowed to $(0.02 - 0.55)$ to stabilize texture refinement. In our optimization pipeline, the “time step” $t \in [0, 1]$ refers to the noise level during the sampling period in the pre-trained 2D diffusion model. Gradient clipping was applied post normalization, starting from an initial value of 1.5 and linearly increasing to 2.0 over 1000 iterations, then remaining constant. This strategy helps maintaining stable gradients throughout the training process. Our approach distinguishes between interaction regions and non-interaction regions by applying a weight $\gamma = 1.5$ to interaction areas while setting other regions to 1. A batch size of 4 was used throughout the training. The distances from the camera ranged from 1.5 to 4.0 during training and were fixed at 4 during evaluation. In terms of view angles, geometric optimization utilized eight fixed views: eight azimuth angles spaced by 45 degrees each and two elevation angles, one at -10° and one at 45° , randomly selecting four views per batch. For texture optimization, the azimuth angles ranged from -180° to 180° , and the elevation angles ranged from -10° to 60° , ensuring comprehensive coverage for rendering. Finally, we rendered 120 views with azimuth intervals of 3° and elevation set to 15° , providing a thorough assessment of the interaction scenes from multiple perspectives.

Recognition of the Interaction Body Part based on GPT-4V. To systematically analyze human-object interactions (HOIs), we establish a fixed part index $\mathcal{M}$ within the human mesh model, which is based on the standard SMPL segmentation protocol. This process involves assigning each vertex to one of $K = 12$ predefined anatomical regions (*e.g.*, head, eyes, spine, hips), ensuring that these semantic annotations remain consistent throughout the optimization process. The structured prior provided by $\mathcal{M}$ serves as a foundational guide for identifying body parts critical to interaction dynamics. Given a text prompt, we utilize GPT-4V [9] to identify the set of key interaction parts $\mathcal{G}$. Once $\mathcal{G}$ is determined, we refer back to our indexed parts $\mathcal{M}$ to extract the corresponding vertices and faces associated with these critical parts. A projection operation is then performed to map these 3D parts onto the 2D image plane, as shown in Fig. 5.

Robustness to GPT-4V Failures and Fallback Strategy. Although GPT-4V generally delivers accurate identification of interaction-critical body parts from text prompts, it can occasionally generate incomplete or erroneous predic-

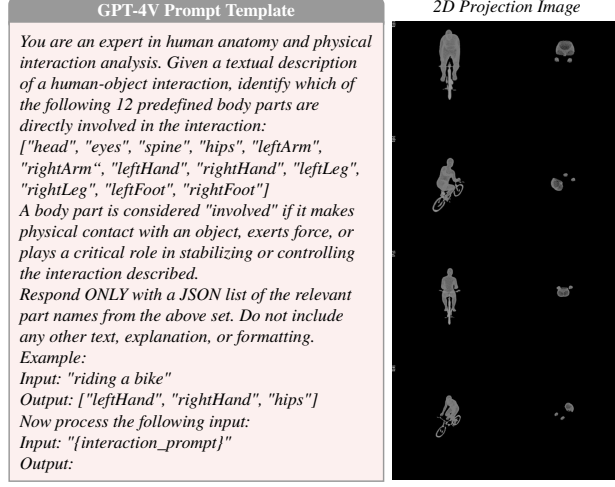


Fig. 5: Prompt template for identifying interaction-critical body parts via GPT-4V [9] and corresponding 2D projection images.

tions (*e.g.*, failing to detect the spine in “carrying a crate” due to ambiguous phrasing). To enhance robustness, our framework integrates a simple yet effective fallback mechanism. Specifically, if the set of key parts $\mathcal{G}$ returned by GPT-4V is empty or contains fewer than two body regions, we automatically revert to a predefined universal interaction prior: $["leftHand", "rightHand", "spine"]$. This prior is motivated by the empirical observation that over 90% of common human-object interactions involve at least one hand and core torso support, as validated on the PROX [3] and BEHAVE [7] datasets. Moreover, even when $\mathcal{G}$ is non-empty but suboptimal, the positions of entities remains well-optimized because the adaptive anchor strategy stabilizes the pose regardless of semantic guidance. In practice, we observed that the qualitative results exhibited no visible artifacts, with interactions maintaining both physically plausible and textural fidelity.

C. Limitations and Future Work

Failure Cases. Although our method demonstrates strong performance in generating physically plausible and texture-faithful 3D human-object interaction scenes, certain limitations persist. We visualize several typical failure cases in Fig. 6. In certain scenarios, excessive density of point clouds on specific body parts poses challenges for optimizing fine human details, often resulting in blurred renderings. As shown in Fig. 6, when the scene involves “sitting on a bed”, scale limitations cause an over-concentration of points on the hands. This high point density constrains texture optimization, thereby limiting the recovery of finger-level details and leading to visually blurred hand regions. Additionally, during



Fig. 6: Failure cases in certain situations.

multi-view texture optimization, human texture refinement tends to be biased when multiple viewing angles are occluded. In Fig. 6, when the scene involves “sitting in a tub chair”, the textures on the lateral of the legs are obscured by the arms and the structure of the seat. This occlusion leads to erroneous texture rendering, causing the optimized regions to inherit appearance characteristics from the chair surface, resulting in textures that visually resemble the surrounding area. These cases underscore the challenges associated with handling severe occlusions and ambiguous object boundaries, highlighting important directions for future research. Notably, these failure cases arise from the inherent difficulties of 3D scene reconstruction under severe occlusion, reflecting common challenges in zero-shot text-to-3D generation rather than limitations of our framework.

Future Work. Although our method enables zero-shot 3D human-object interaction synthesis without relying on any 3D training data, recent two-stage pipelines, such as Flux for high-fidelity 2D generation followed by Trellis for 2D-to-3D reconstruction, demonstrate compelling realism, albeit at the cost of requiring large-scale image-3D paired datasets. This highlights a fundamental trade-off between data efficiency and output fidelity. In future work, we aim to bridge this gap by exploring hybrid strategies that retain our zero-shot capability while incorporating lightweight, self-supervised geometric priors or distillation from pre-trained 2D diffusion models. Ultimately, we envision a framework that achieves the visual quality of data-driven approaches without sacrificing the generalization and flexibility of zero-shot synthesis.

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